

2025-12-04 - Servidio (16)

Previous lecture: [2025-11-28 - Servidio \(15\)](#)

Next lecture: [2025-12-05 - Valentini \(17\)](#)

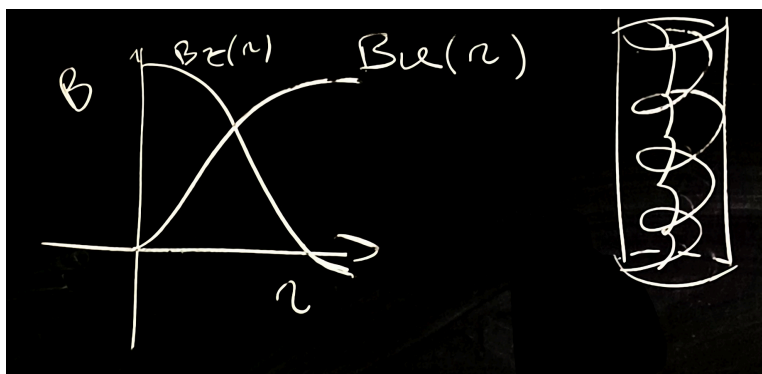
Recap

Force-free equilibria

$$\nabla \times \bar{B} = \alpha \bar{B}$$

in cylindrical geometry

PIC - Bz Bth



We find equilibrium in system similar to solar corona to obtain coronal loops



Force free equilibria in cartesian geometry

Recall $\nabla \times \bar{B} = \mu \bar{B}$.

2.5D, fields depend on x and y , but they can have an out-of-plane component

(eg. fields can have $A_z(x, y)$)

So we have

$$\partial_z = 0, \quad \bar{A} = [A_x(x, y), A_y(x, y), A_z(x, y)]$$

We work in $x - y$ plane

We use the Helmholtz decomposition (for solenoidal field)

$$\bar{B} = \nabla \Psi \times \hat{z} + B_z \hat{z} \quad \text{and} \quad \bar{B}_{\text{planar}} = \nabla \Psi \times \hat{z} = \begin{cases} B_x = \partial_y \Psi \\ B_y = -\partial_x \Psi \end{cases}$$

and

$$\nabla \times \bar{a} = \nabla \times (\Psi \hat{z}) = \nabla \Psi \times \hat{z}$$

so with this we work from

$$(B_x, B_y, B_z) \rightarrow (\Psi, B_z)$$

we find J_z as ($\Psi = \Psi(x, y)$)

$$J_z = \nabla \times \bar{B}_{\text{planar}} = -\nabla^2 \Psi$$

so

$$\begin{aligned} \mu \bar{B} &= \nabla \times \bar{B} = \\ &= \nabla \times (\nabla \Psi \times \hat{z}) + \nabla \times (B_z \hat{z}) = \\ &= -\nabla^2 \Psi \hat{z} + \nabla \times (B_z \hat{z}) \end{aligned}$$

and

$$\begin{cases} \mu B_x = \partial_y B_z \\ \mu B_y = -\partial_x B_z \\ \mu B_z = -\nabla^2 \Psi \end{cases}$$

we rewrite this using the derivatives of Ψ

$$\begin{cases} \mu \partial_y \Psi = \partial_y B_z \\ -\mu \partial_x \Psi = -\partial_x B_z \\ \mu B_z = -\nabla^2 \Psi \end{cases}$$

from the first two we have

$$\begin{cases} \partial_y (\mu \Psi - B_z) = 0 \\ \partial_x (\mu \Psi - B_z) = 0 \end{cases} \implies B_z = \mu \Psi$$

combining this with the third

$$\nabla^2 \Psi + \mu^2 \Psi = 0$$

which is known as **Helmholtz equation**.

We solve it

$$\frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} + \mu^2 \Psi = 0$$

Periodic boundaries (Fourier decomposition only on x)

$$\Psi(x, y) = \sum_l \tilde{\Psi}(l, y) e^{ilx}$$

so the Helmholtz equation becomes

$$\sum_l (-l^2 \tilde{\Psi} e^{ilx}) + \sum_l \frac{\partial^2 \tilde{\Psi}}{\partial y^2} e^{ilx} + \sum_l \mu^2 \tilde{\Psi} e^{ilx} = 0$$

we multiply by e^{-ikx} in order to eliminate the sum and focus on one wavenumber at a time
(we could use the orthogonality of the base and avoid this passage)

$$\begin{aligned} & \sum_l (-l^2 \tilde{\Psi}(l, y) e^{i(l-k)x}) + \sum_l \frac{\partial^2 \tilde{\Psi}(l, y)}{\partial y^2} e^{i(l-k)x} + \sum_l \mu^2 \tilde{\Psi}(l, y) e^{i(l-k)x} = 0 \\ \implies & \sum_l (-l^2 \tilde{\Psi}(l, y) \delta_{l,k}) + \sum_l \frac{\partial^2 \tilde{\Psi}(l, y)}{\partial y^2} \delta_{l,k} + \sum_l \mu^2 \tilde{\Psi}(l, y) \delta_{l,k} = 0 \\ \implies & -k^2 \tilde{\Psi}(k, y) + \frac{\partial^2 \tilde{\Psi}(k, y)}{\partial y^2} + \mu^2 \tilde{\Psi}(k, y) = 0 \end{aligned}$$

so

$$\frac{\partial^2 \tilde{\Psi}(k, y)}{\partial y^2} + (\mu^2 - k^2) \tilde{\Psi}(k, y) = 0$$

At infinity we don't have magnetic field and at $y = 0$ we have a finite magnetic field

$$\tilde{\Psi}(k, y) \xrightarrow{y \rightarrow +\infty} 0; \quad \tilde{\Psi}(k, y = 0) = \tilde{\Psi}_0(k)$$

On large scales (small vortices die first) we have only k_0 .

Using only k_0 and using the previous boundary condition we have the following solution
(procedure left as exercise)

$$\Psi(x, y) = e^{-y\sqrt{k_0^2 - \mu^2}} \cos(k_0 x)$$

We verify that it's a solution

$$\begin{aligned} \frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} + \mu^2 \Psi &= -k_0^2 e^{-y\sqrt{k_0^2 - \mu^2}} \cos(k_0 x) + \\ &+ (k_0^2 - \mu^2) e^{-y\sqrt{k_0^2 - \mu^2}} \cos(k_0 x) + \mu^2 e^{-y\sqrt{k_0^2 - \mu^2}} \cos(k_0 x) = 0 \end{aligned}$$

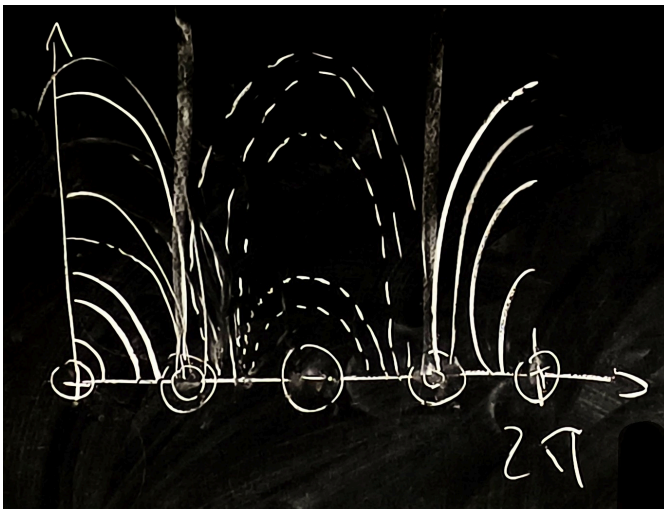
We plot it

$$x \in [0, 2\pi] = [0, L_0] \implies k = 1 = \frac{2\pi}{L_0}$$

$$\mu = \frac{1}{\sqrt{2}}$$

(isolines, continuous $\rightarrow$ positive, dashed $\rightarrow$ negative)

PIC isolines



we can build $\bar{B}$ vectors using $\nabla\Psi \times \hat{z}$

PIC b vec



We can use $B_z = \mu\Psi$ to find B_z

it is exiting from the picture where Ψ is positive and entering where Ψ is negative.

These structure don't last for long.

To sum up

$$\nabla \times \bar{B} = \mu\bar{B}$$

in clindrical coordinates we have

$$\begin{cases} B_z \sim J_0(\alpha r) \\ B_\theta \sim J_1(\alpha r) \end{cases}$$

in cartesian coordinates we have

$$\bar{B} = \nabla\Psi \times \hat{z} + \mu\Psi\hat{z} \quad \implies \quad \Psi \sim e^{-y\sqrt{k_0^2 - \mu^2}} \cos(k_0 x)$$

Bidimensional MHD

We start from

$$\begin{cases} \frac{\partial \bar{U}}{\partial t} = -\bar{U} \cdot \nabla \bar{U} - \nabla P + \bar{B} \cdot \nabla \bar{B} + \nu \nabla^2 \bar{U} \\ \frac{\partial \bar{B}}{\partial t} = \nabla \times [\bar{U} \times \bar{B} - \eta \nabla \times \bar{B}] \end{cases}$$

We assume

$$\nu = 0, \quad \eta = 0, \quad \bar{U} = 0$$

so

$$\begin{cases} \frac{\partial \bar{U}}{\partial t} = -\nabla P + \bar{B} \cdot \nabla \bar{B} \\ \frac{\partial \bar{B}}{\partial t} = 0 \end{cases}$$

we apply the curl to the first equation and we take out the curl in the second ($\bar{B} = \nabla \Psi \times \hat{z}$)
(the pressure term vanish bcause the rotor of a gradient is null)

$$\begin{cases} 0 = \bar{B} \cdot \nabla j_z \\ \frac{\partial \Psi}{\partial t} = 0 \end{cases}$$

The full equation would be the following, but we don't need for this calculation

$$\begin{cases} \frac{\partial \omega_z}{\partial t} = -\bar{U} \cdot \nabla \omega_z + \bar{B} \cdot \nabla j_z + \nu \nabla^2 \bar{\omega} \\ \frac{\partial \Psi}{\partial t} = -\bar{U} \cdot \nabla \Psi + \eta \nabla^2 \Psi \end{cases}$$

We now continue from

$$\bar{B} \cdot \nabla j_z = 0$$

we recall that

$$\bar{B} = \nabla \Psi \times \hat{z}, \quad j_z = -\nabla^2 \Psi$$

after we substitute we get

$$(\nabla \Psi \times \hat{z}) \cdot \nabla (\nabla^2 \Psi) = 0$$

expanding the scalar product

$$-\partial_y \Psi \cdot \partial_x (\nabla^2 \Psi) + \partial_x \Psi \cdot \partial_y (\nabla^2 \Psi) = 0$$

We choose j_z as a function of Ψ : $j_z = f(\Psi)$. We can do this because it's compatible with the condition $j_z = -\nabla^2 \Psi$. Then we get

$$\begin{aligned} & -\partial_y \Psi \cdot \partial_x (f(\Psi)) + \partial_x \Psi \cdot \partial_y (f(\Psi)) = 0 \\ \implies & -\partial_y \Psi \cdot \frac{\partial f}{\partial \Psi} \cdot \partial_x \Psi + \partial_x \Psi \cdot \frac{\partial f}{\partial \Psi} \cdot \partial_y \Psi = 0 \\ \implies & \frac{\partial f}{\partial \Psi} [-\partial_y \Psi \cdot \partial_x \Psi + \partial_x \Psi \cdot \partial_y \Psi] = 0 \end{aligned}$$

so we go from a wide class of function to a narrower.

we restrict again by using force free

$$j_z = f(\Psi) = \mu^2 \Psi \quad \implies \quad \nabla^2 \Psi + \mu^2 \Psi = 0$$

another interesting restriction is

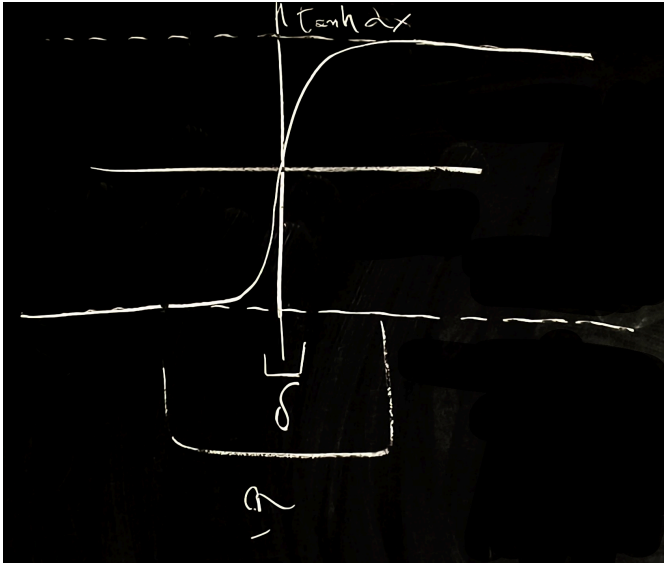
$$j_z = f(\Psi) = e^{\alpha\Psi} \quad \Rightarrow \quad \nabla^2\Psi + \alpha e^{\beta\Psi} = 0$$

which is very realistic in plasma

One famous solution is

$$j_z(x) = \frac{2\alpha^2}{\cosh^2 \alpha x} \quad \Rightarrow \quad B_y = 2\alpha \tanh \alpha x$$

PIC of tanh



mag field is negative on one side and positive on the other, smooth discontinuity

tanh changes sign at $x = 0$

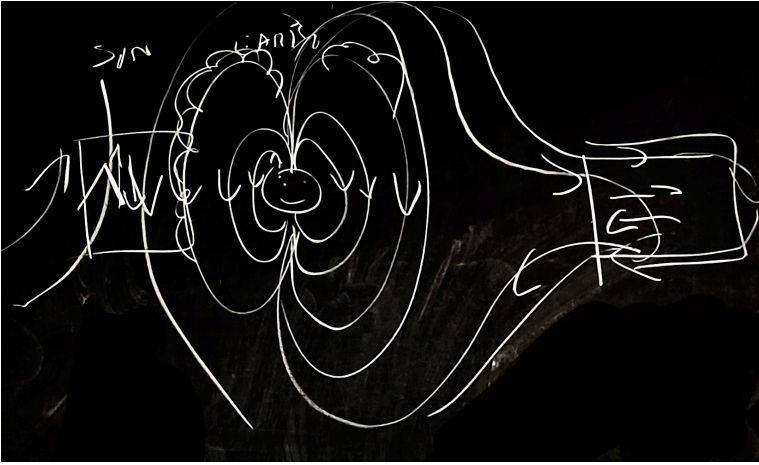
it is an unstable point

Examples

PIC cme inst.

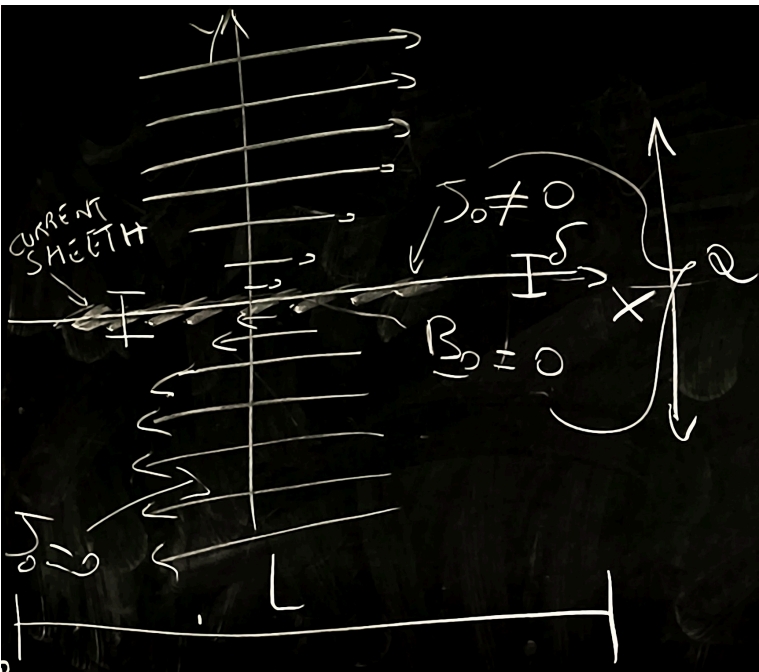


PIC sw and earth mag tail



Tearing instability

PIC b0

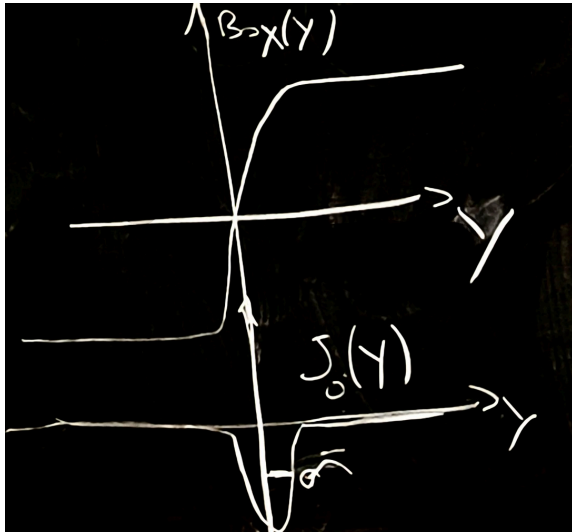


In the system we have

$$\bar{B}_0 = B_{0x}(y)\hat{x} = B_0 \tanh\left(\frac{y}{a}\right)\hat{x}$$

$$\bar{J}_0 = J_0(y)\hat{z} = -\partial_y B_{0x} \simeq -\frac{1}{\cosh^2\left(\frac{y}{a}\right)}\hat{z}$$

PIC b0x j0y



current sheet $\rightarrow$ curr = derivative of B

The current sheet is the sheet close to $y = 0$, of thickness δ . Here $J_0 \neq 0$

a typical length of magnetic field flip

L typical size of system (large)

δ typical thickness of current sheet

$$\delta \ll a \ll L$$

We perturb the system with incompressible fluctuations

$$\begin{cases} \bar{v}_1 = v_x(x, y, t)\hat{x} + v_y(x, y, t)\hat{y} \\ \bar{b}_1 = b_x(x, y, t)\hat{x} + b_y(x, y, t)\hat{y} \end{cases}$$

with

$$|\bar{v}_1|, |\bar{b}_1| \rightarrow 0, \quad \nu = 0, \quad \eta \neq 0$$

so we have a real plasma with resistivity

$$\begin{cases} \frac{\partial \bar{V}_1}{\partial t} = -\cancel{\bar{V}_1 \cdot \nabla \bar{V}_1} - \nabla P_1 + \bar{J}_0 \times \bar{b}_1 + \bar{J}_1 \times \bar{b}_0 + \cancel{\bar{J}_0 \times \bar{b}_0} + \cancel{\bar{J}_1 \times \bar{b}_1} \\ \frac{\partial \bar{b}_1}{\partial t} = \nabla \times (\bar{V}_1 \times \bar{B}_0) + \cancel{\nabla \times (\bar{V}_1 \times \bar{b}_1)} + \eta \nabla^2 \bar{b}_1 \end{cases}$$

which gets simplified to

$$\begin{cases} \frac{\partial \bar{V}_1}{\partial t} = -\nabla P_1 + \bar{J}_0 \times \bar{b}_1 + \bar{J}_1 \times \bar{b}_0 \\ \frac{\partial \bar{b}_1}{\partial t} = \nabla \times (\bar{V}_1 \times \bar{B}_0) + \eta \nabla^2 \bar{b}_1 \\ \nabla \cdot \bar{V}_1 = \nabla \cdot \bar{b}_1 = 0 \end{cases}$$

Hypothesis (valid only in the very first stage, before nonlinearity)

$$\begin{cases} \bar{V}_1 = \bar{V}(y)e^{\gamma t + i k x} \\ \bar{b}_1 = \bar{b}(y)e^{\gamma t + i k x} \end{cases}$$

we want to find the γ

we use the notation $\partial_y f = f'$

$$\nabla \cdot \bar{b}_1 = 0 \quad \implies \quad ikb_x(y)e^{\gamma t + ikx} + b'_y(y)e^{\gamma t + ikx} = 0$$

so

$$b_x(y) = i \frac{b'_y(y)}{k}$$

and, following the same reasoning

$$\nabla \cdot \bar{v}_1 = 0 \quad \implies \quad v_x(y) = i \frac{v'_y(y)}{k}$$

now we take the curl of the first equation to get rid of the pressure

$$\frac{\partial \nabla \times \bar{v}_1}{\partial t} = \nabla \times [\bar{J}_0 \times \bar{b}_1 + \bar{J}_1 \times \bar{B}_0]$$

and we only consider the $\hat{z}$ component.

considering also

$$\begin{aligned} \bar{J}_0 &= J_0 \hat{z} = -\partial_y B_{0x} \hat{z} = -B'_{0x} \hat{z} \\ \bar{J}_0 \times \bar{b}_1 + \bar{J}_1 \times \bar{B}_0 &= \begin{pmatrix} b_y B'_{0x} \\ -b_x B'_{0x} + B_{0x} [\partial_x b_y - \partial_y b_x] \\ 0 \end{pmatrix} \end{aligned}$$

we need to evaluate the rotor of it

and we get (recall $b_x(y) = \frac{i}{k} b'_y(y)$)

$$\begin{aligned} \frac{\partial \nabla \times \bar{v}_1}{\partial t} \cdot \hat{z} &= \nabla \times [\bar{J}_0 \times \bar{b}_1 + \bar{J}_1 \times \bar{B}_0] \cdot \hat{z} = \\ &= \partial_x [-b_x B'_{0x} + B_{0x} (\partial_x b_y - \partial_y b_x)] - \partial_y [b_y B'_{0x}] = \\ &= \left\{ \cancel{-ikb_x B'_{0x}} - B_{0x} b_y k^2 - b'_x ik B_{0x} - \cancel{b'_y B'_{0x}} - b_y B''_{0x} \right\} e^{\gamma t + ikx} \\ &= \left\{ -B_{0x} b_y k^2 - b'_x ik B_{0x} - b_y B''_{0x} \right\} e^{\gamma t + ikx} = \\ &= \left\{ B_{0x} (-b'_x ik - b_y k^2) - b_y B''_{0x} \right\} e^{\gamma t + ikx} = \\ &= \left\{ B_{0x} (b''_y - k^2 b_y) - b_y B''_{0x} \right\} e^{\gamma t + ikx} \end{aligned}$$

the LHS is (recall $v_x(y) = \frac{i}{k} v'_y(y)$)

$$\begin{aligned} \frac{\partial \nabla \times \bar{v}_1}{\partial t} \cdot \hat{z} &= \frac{\partial}{\partial t} \{ \partial_x [V_y(y) e^{\gamma t + ikx}] - \partial_y [V_x(y) e^{\gamma t + ikx}] \} = \\ &= \frac{\partial}{\partial t} \{ [ikV_y - V'_x] e^{\gamma t + ikx} \} = \\ &= \gamma \left(ikV_y - \frac{i}{k} V''_y \right) e^{\gamma t + ikx} = \\ &= -\frac{i\gamma}{k} (V''_y - k^2 V_y) e^{\gamma t + ikx} \end{aligned}$$

and combining them we have

$$-\frac{i\gamma}{k}(V_y'' - k^2 V_y) = B_{0x}(b_y'' - k^2 b_y) - b_y B_{0x}''$$

so $(-i = i^{-1})$

$$\gamma(V_y'' - k^2 V_y) = ik[B_{0x}(b_y k^2 - ik b_x') - b_y B_{0x}'']$$

The following part is explained better in the next lesson.

And for the magnetic field

$$\frac{\partial \bar{b}_1}{\partial t} = \nabla \times (\bar{V}_1 \times \bar{B}_0) + \eta \nabla^2 \bar{b}_1 \quad \implies \quad \gamma b_y = ik B_{0x} V_y + \eta(b_y'' - k^2 b_y)$$

now if we change the variables names to

$$\begin{cases} V = v_y \\ b = b_y \end{cases}$$

so we have

$$\begin{cases} \gamma(V'' - k^2 V) = ik[B_{0x}(b'' - k^2 b) - b B_{0x}'] \\ \gamma b = ik B_{0x} V + \eta(b'' - k^2 b) \end{cases}$$

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